

=== GNB (1780742698) ===

```
# This configuration file example shows how to configure the srsRAN Project gNB to allow srsUE to connect to
it.
# This specific example uses ZMQ in place of a USRP for the RF-frontend, and creates an FDD cell with 10 MHz
bandwidth.
# To run the srsRAN Project gNB with this config, use the following command:
#   sudo ./gnb -c gnb_zmq.yaml

cu_cp:
  amf:
    addr: 10.53.1.2          # The address or hostname of the AMF.
    port: 38412
    bind_addr: 10.53.1.1    # A local IP that the gNB binds to for traffic from the AMF.
    supported_tracking_areas:
      - tac: 7
        plmn_list:
          - plmn: "00101"
            tai_slice_support_list:
              - sst: 1
    inactivity_timer: 7200  # Sets the UE/PDU Session/DRB inactivity timer to 7200 seconds. Supported:
[1 - 7200].

ru_sdr:
  device_driver: zmq        # The RF driver name.
  device_args: tx_port=tcp://127.0.0.1:2000,rx_port=tcp://127.0.0.1:2001,base_srate=11.52e6 # Optionally pass
arguments to the selected RF driver.
  srate: 11.52             # RF sample rate might need to be adjusted according to selected bandwidth.
  tx_gain: 75              # Transmit gain of the RF might need to adjusted to the given situation.
  rx_gain: 75              # Receive gain of the RF might need to adjusted to the given situation.

cell_cfg:
  dl_arfcn: 368500         # ARFCN of the downlink carrier (center frequency).
  band: 3                  # The NR band.
  channel_bandwidth_MHz: 20 # Bandwith in MHz. Number of PRBs will be automatically derived.
  common_scs: 15           # Subcarrier spacing in kHz used for data.
  ssb_scs: 15              # Subcarrier spacing in kHz used for SSB.
  plmn: "00101"            # PLMN broadcasted by the gNB.
  tac: 7                   # Tracking area code (needs to match the core configuration).
  pdcch:
    common:
      ss0_index: 0         # Set search space zero index to match srsUE capabilities
      coresets0_index: 6   # Set search CORESET Zero index to match srsUE capabilities
    dedicated:
      ss2_type: common     # Search Space type, has to be set to common
      dci_format_0_1_and_1_1: false # Set correct DCI format (fallback)
  prach:
    prach_config_index: 1  # Sets PRACH config to match what is expected by srsUE
  pdsch:
    mcs_table: qam64       # Sets PDSCH MCS to 64 QAM
  pusch:
    mcs_table: qam64       # Sets PUSCH MCS to 64 QAM

log:
  filename: /tmp/gnb.log   # Path of the log file.
  all_level: info          # Logging level applied to all layers.
  hex_max_size: 0

pcap:
  mac_enable: false        # Set to true to enable MAC-layer PCAPs.
  mac_filename: /tmp/gnb_mac.pcap # Path where the MAC PCAP is stored.
  ngap_enable: false       # Set to true to enable NGAP PCAPs.
  ngap_filename: /tmp/gnb_ngap.pcap # Path where the NGAP PCAP is stored.
  e2ap_enable: true        # Set to true to enable E2AP PCAPs.
  e2ap_du_filename: /tmp/gnb_du_e2ap.pcap # Path where the DU E2AP PCAP is stored.
  e2ap_cu_cp_filename: /tmp/gnb_cu_cp_e2ap.pcap # Path where the CU-CP E2AP PCAP is stored.
```

e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap # Path where the CU-UP E2AP PCAP is stored.

#e2:

```
# enable_du_e2: true           # Enable DU E2 agent (one for each DU instance)
# e2sm_kpm_enabled: true       # Enable KPM service module
# e2sm_rc_enabled: true        # Enable RC service module
# addr: 127.0.0.1              # RIC IP address
# bind_addr: 127.0.0.100       # A local IP that the E2 agent binds to for traffic from the RIC. ONLY
                                # required if running the RIC on a separate machine.
# port: 36421                  # RIC port
```

metrics:

layers:

```
    enable_rlc: true           # Enable RLC metrics
    enable_sched: true         # Enable DU scheduler metrics
periodicity:
    du_report_period: 1000     # Set DU statistics report period to 1s
```

=== UE (1780742698) ===

```
[rf]
freq_offset = 0
tx_gain = 50
rx_gain = 40
srate = 11.52e6
nof_antennas = 1

device_name = zmq
device_args = tx_port=tcp://127.0.0.1:2001,rx_port=tcp://127.0.0.1:2000,base_srate=11.52e6

[rat.eutra]
dl_eurfcn = 2850
nof_carriers = 0

[rat.nr]
bands = 3
nof_carriers = 1

max_nof_prb = 106
nof_prb = 106

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc = 112233445566778899aabbccddeeff00
k = 00112233445566778899aabbccddeeff
imsi = 001010000000001

[rrc]
release = 15
ue_category = 4

[nas]
apn = srsapn
apn_protocol = ipv4

[gw]
netns = ue1
ip_devname = tun_srsue
ip_netmask = 255.255.255.0

[gui]
enable = false
```

=== DOCKER (1780742698) ===

```
#
# Copyright 2021-2026 Software Radio Systems Limited
#
# This file is part of srsRAN
#
# srsRAN is free software: you can redistribute it and/or modify
# it under the terms of the GNU Affero General Public License as
# published by the Free Software Foundation, either version 3 of
# the License, or (at your option) any later version.
#
# srsRAN is distributed in the hope that it will be useful,
# but WITHOUT ANY WARRANTY; without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the
# GNU Affero General Public License for more details.
#
# A copy of the GNU Affero General Public License can be found in
# the LICENSE file in the top-level directory of this distribution
# and at http://www.gnu.org/licenses/.
#
```

services:

```
5gc:
  container_name: open5gs_5gc
  build:
    context: open5gs
    target: open5gs
    args:
      OS_VERSION: "22.04"
      OPEN5GS_VERSION: "v2.7.0"
  environment:
    - MONGODB_IP=127.0.0.1
    - OPEN5GS_IP=10.53.1.2
    - UE_IP_BASE=10.45.0
    - UPF_ADVERTISE_IP=10.53.1.2
    - DEBUG=false
```

SUBSCRIBER_DB=001010000000001,00112233445566778899aabbccddeeff,opc,112233445566778899aabbccddeeff00,8000,9,10.45.1.2

```
-
  - NETWORK_NAME_FULL=srsRAN
  - NETWORK_NAME_SHORT=srsRAN
  - TZ=Europe/Madrid
privileged: true
ports:
  - "9898:9999/tcp"
# Uncomment port to use the 5gc from outside the docker network
# - "38412:38412/sctp"
# - "2152:2152/udp"
command: 5gc -c open5gs-5gc.yml
healthcheck:
  test: [ "CMD-SHELL", "nc -z 127.0.0.20 7777" ]
  interval: 3s
  timeout: 1s
  retries: 60
networks:
  ran:
    ipv4_address: ${OPEN5GS_IP:-10.53.1.2}
```

```
gnb:
  container_name: srsran_gnb
# Build info
image: srsran/gnb
build:
  context: ..
  dockerfile: docker/Dockerfile
```

```

args:
  OS_VERSION: "24.04"
# privileged mode is required only for accessing usb devices
privileged: true
# Extra capabilities always required
cap_add:
  - SYS_NICE
  - CAP_SYS_PTRACE
volumes:
  # Access USB to use some SDRs
  - /dev/bus/usb/:/dev/bus/usb/
  # Sharing images between the host and the pod.
  # It's also possible to download the images inside the pod
  - /usr/share/uhd/images:/usr/share/uhd/images
  # Save logs and more into gnb-storage
  - gnb-storage:/tmp
# It creates a file/folder into /config_name inside the container
# Its content would be the value of the file used to create the config
configs:
  - gnb_config.yml
  - gnb_compose_config.yml
# Customize your desired network mode.
# current netowrk configuration creastes a private netwoek with both containers attached
# An alterantive would be `network: host`. That would expose your host network into the container. It's
the easiest to use if the 5gc is not in your PC
networks:
  ran:
    ipv4_address: ${GNB_IP:-10.53.1.3}
  metrics:
    ipv4_address: 172.19.1.3
# Start GNB container after 5gc is up and running
depends_on:
  5gc:
    condition: service_healthy
# Command to run into the final container
command: gnb -c /gnb_config.yml -c /gnb_compose_config.yml

configs:
  gnb_config.yml:
    content: |
      # This configuration file example shows how to configure the srsRAN Project gNB to allow srsUE to connect
      to it.
      # This specific example uses ZMQ in place of a USRP for the RF-frontend, and creates an FDD cell with 10
      MHz bandwidth.
      # To run the srsRAN Project gNB with this config, use the following command:
      #   sudo ./gnb -c gnb_zmq.yaml

  cu_cp:
    amf:
      addr: 10.53.1.2          # The address or hostname of the AMF.
      port: 38412
      bind_addr: 10.53.1.1    # A local IP that the gNB binds to for traffic from the AMF.
      supported_tracking_areas:
        - tac: 7
          plmn_list:
            - plmn: "00101"
            tai_slice_support_list:
              - sst: 1
          inactivity_timer: 7200 # Sets the UE/PDU Session/DRB inactivity timer to 7200 seconds.
      Supported: [1 - 7200].

  ru_sdr:
    device_driver: zmq          # The RF driver name.
    device_args: tx_port=tcp://127.0.0.1:2000,rx_port=tcp://127.0.0.1:2001,base_srate=11.52e6 # Optionally
    pass arguments to the selected RF driver.
    srate: 11.52              # RF sample rate might need to be adjusted according to selected
    bandwidth.
    tx_gain: 75               # Transmit gain of the RF might need to adjusted to the given

```

```

situation.
    rx_gain: 75                                # Receive gain of the RF might need to adjusted to the given
situation.

cell_cfg:
    dl_arfcn: 368500                            # ARFCN of the downlink carrier (center frequency).
    band: 3                                    # The NR band.
    channel_bandwidth_MHz: 20                  # Bandwith in MHz. Number of PRBs will be automatically derived.
    common_scs: 15                             # Subcarrier spacing in kHz used for data.
    ssb_scs: 15                               # Subcarrier spacing in kHz used for SSB.
    plmn: "00101"                             # PLMN broadcasted by the gNB.
    tac: 7                                    # Tracking area code (needs to match the core configuration).
    pdccch:
        common:
            ss0_index: 0                        # Set search space zero index to match srsUE capabilities
            coresets0_index: 6                 # Set search CORESET Zero index to match srsUE capabilities
        dedicated:
            ss2_type: common                    # Search Space type, has to be set to common
            dci_format_0_1_and_1_1: false      # Set correct DCI format (fallback)
    prach:
        prach_config_index: 1                  # Sets PRACH config to match what is expected by srsUE
    pdsch:
        mcs_table: qam64                       # Sets PDSCH MCS to 64 QAM
    pusch:
        mcs_table: qam64                       # Sets PUSCH MCS to 64 QAM

log:
    filename: /tmp/gnb.log                     # Path of the log file.
    all_level: info                           # Logging level applied to all layers.
    hex_max_size: 0

pcap:
    mac_enable: false                         # Set to true to enable MAC-layer PCAPs.
    mac_filename: /tmp/gnb_mac.pcap           # Path where the MAC PCAP is stored.
    ngap_enable: false                       # Set to true to enable NGAP PCAPs.
    ngap_filename: /tmp/gnb_ngap.pcap         # Path where the NGAP PCAP is stored.
    e2ap_enable: true                        # Set to true to enable E2AP PCAPs.
    e2ap_du_filename: /tmp/gnb_du_e2ap.pcap   # Path where the DU E2AP PCAP is stored.
    e2ap_cu_cp_filename: /tmp/gnb_cu_cp_e2ap.pcap # Path where the CU-CP E2AP PCAP is stored.
    e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap # Path where the CU-UP E2AP PCAP is stored.

#e2:
# enable_du_e2: true                         # Enable DU E2 agent (one for each DU instance)
# e2sm_kpm_enabled: true                     # Enable KPM service module
# e2sm_rc_enabled: true                      # Enable RC service module
# addr: 127.0.0.1                           # RIC IP address
# bind_addr: 127.0.0.100                     # A local IP that the E2 agent binds to for traffic from the RIC.
ONLY required if running the RIC on a separate machine.
# port: 36421                               # RIC port

metrics:
    layers:
        enable_rlc: true                     # Enable RLC metrics
        enable_sched: true                   # Enable DU scheduler metrics
    periodicity:
        du_report_period: 1000               # Set DU statistics report period to 1s
gnb_compose_config.yml:
content: |
cu_cp:
amf:
    addr: ${OPEN5GS_IP:-10.53.1.2}
    bind_addr: 0.0.0.0
metrics:
    autostart_stdout_metrics: true
    enable_json: true
remote_control:
    bind_addr: 0.0.0.0
    enabled: true

```

```
volumes:  
  gnb-storage:
```

```
networks:  
  ran:  
    ipam:  
      driver: default  
      config:  
        - subnet: 10.53.1.0/24  
metrics:  
  ipam:  
    driver: default  
    config:  
      - subnet: 172.19.1.0/24
```

=== NOTAS (1780742698) ===

- Se ha aplicado la REGLA 1 (Coherencia PLMN): MCC "001" y MNC "01" (plmn "00101") se usan en gNB, UE e IMSI.
- Se ha aplicado la REGLA 2 (Coherencia de frecuencias): dl_arfcn 368500 está en el rango de Banda 3.
- Se ha aplicado la REGLA 3 (Coherencia SCS): Añadido 'ssb_scs: 15' para que coincida con 'common_scs'.
- Se ha aplicado la REGLA 4 (Puertos ZMQ cruzados): Los puertos ZMQ están correctamente cruzados entre gNB y UE.
- Se ha aplicado la REGLA 5 (bind_addr): gNB standalone usa 10.53.1.1; gnb_compose_config usa 0.0.0.0.
- Se ha aplicado la REGLA 6 (Bloque ru_sdr): Copiado directamente del CAG sin modificaciones.
- Se ha aplicado la REGLA 7 (Coherencia IMSI y SUBSCRIBER_DB): IMSI, K y OPC son consistentes en UE y SUBSCRIBER_DB, incluyendo el orden de K y OPC.
- El campo `channel\_bandwidth\_MHz` en `gnb\_zmq.yaml` se ha fijado a 20 MHz (resolviendo la duplicidad del template).